

Section A: Scenario-Based Concept Application

1. Explain the operational factors that justify selecting a specific Linux distribution for a startup's internal tool hosting environment.

Ans :- Operational factors to consider:

- `ssh-keygen -t rsa -b 4096` Security Updates: Regular patches protect servers from vulnerabilities.
- Community Support: Large communities provide documentation and troubleshooting assistance.
- Package Availability: Easy installation of web services like Apache, MySQL, and PHP.
- Cost: Ubuntu is free, while RHEL provides enterprise support at a subscription cost.
- Performance: Lightweight distributions consume fewer resources.
- Compatibility: Support for cloud platforms and automation tools.

For startups, Ubuntu Server LTS is often preferred because it is free, stable, and well-supported.

2. Explain the implementation of SSH Key-based Authentication to satisfy a security requirement where password logins must be disabled for developer access.

Ans :- Generate SSH Key Pair

- `ssh-keygen -t rsa -b 4096`

Copy Public Key to Server

- `ssh-copy-id user@server-ip`

Edit SSH Configuration

- `sudo nano /etc/ssh/sshd_config`

Set:

- `PubkeyAuthentication yes`
- `PasswordAuthentication no`
- `PermitRootLogin no`

Restart SSH:

- `sudo systemctl restart sshd`

Benefits:

- Prevents brute-force password attacks.
- Stronger authentication.
- Supports automation scripts securely.

3. Explain the configuration of Linux Permissions and Ownership for the /var/www directory to support a multi-user environment.

Ans :- Linux Permissions and Ownership for /var/www

Create group:

- `sudo groupadd webadmins`

Add users:

- `sudo usermod -aG webadmins user1`

Change ownership:

- `sudo chown -R root:webadmins /var/www`

Set permissions:

- `sudo chmod -R 775 /var/www`

Set SGID:

- `sudo chmod g+s /var/www`

Benefits:

- Multiple developers can manage website files.
- New files inherit group ownership.
- Restricts unauthorized access.

4. Explain how Logical Volume Management (LVM) facilitates dynamic storage expansion to prevent service downtime when disk space is depleted.

Ans :- Advantages:

- Online storage expansion.
- Flexible partition management.
- No downtime during disk expansion.

Example:-

Add new disk:

- `pvcreate /dev/sdb`
- `vgextend vg_data /dev/sdb`
- `lvextend -l +100%FREE /dev/vg_data/lv_web`
- `resize2fs /dev/vg_data/lv_web`

Benefits:

- Prevents service outages due to insufficient disk space.
- Easy future scalability.

5. Explain the troubleshooting process for diagnosing and resolving service blocks caused by SELinux security policies when a website becomes inaccessible.

Ans :- Troubleshooting SELinux Issues

Check SELinux status:

- `Getenforce`

View logs:

- `sudo ausearch -m avc`

Check audit log:

- `sudo cat /var/log/audit/audit.log`

Temporarily disable:

- `sudo setenforce 0`

If website works after disabling SELinux:

- `sudo restorecon -Rv /var/www/html`
- `sudo setsebool -P httpd_can_network_connect on`

Re-enable:

- `sudo setenforce 1`

6. Explain the safe configuration of Cron Jobs to ensure critical daily backup tasks execute automatically at a scheduled time.

Ans :- Safe Cron Job Configuration

Edit cron:

- `crontab -e`

Daily backup at 2 AM:

- `0 2 * * * tar -czf /backup/www-$(date +%F).tar.gz /var/www`

Redirect logs:

- `0 2 * * * tar -czf /backup/www-$(date +%F).tar.gz /var/www >> /var/log/backup.log 2>&1`

Benefits:

- Automatic backups.
- Logging for troubleshooting.
- Reduces risk of data loss.

Section B: Practical Application

1. System Identity: Set the system hostname to startup-web-01 and verify the change via CLI.

Ans :- Set Hostname

- `sudo hostnamectl set-hostname startup-web-01`
- `hostnamectl`
- `hostname`

2. Network Setup: Configure a static IP address on the primary interface and restart the networking service.

Ans :- Configure Static IP

- `sudo nmcli connection modify ens33 \`
- `ipv4.addresses 192.168.1.100/24 \`
- `ipv4.gateway 192.168.1.1 \`
- `ipv4.dns "8.8.8.8 8.8.4.4" \`
- `ipv4.method manual`

3. Secure Access: Enable the SSH service and ensure it starts automatically on system boot.

Ans :- For CentOS

- `sudo yum install openssh-server -y`
- `sudo systemctl enable sshd`
- `sudo systemctl start sshd`

4. Access Control: Create a group named webadmins and a user dev-ops with restricted sudo privileges.

Ans :- Create Group and User

- `sudo groupadd webadmins`
- `sudo useradd -m -G webadmins dev-ops`
- `sudo passwd dev-ops`

Restricted sudo:

- `sudo visudo`

Add:

- `dev-ops ALL=(ALL) /bin/systemctl,/usr/bin/journalctl`

5. Web Hosting: Install the Apache service and deploy a custom index.html file to /var/www/html.

Ans :- Install Apache and Deploy Website

- `sudo yum install httpd -y`

Start service:

- `sudo systemctl enable apache2`
- `sudo systemctl start apache2`

Create webpage:

- `echo "<h1>Welcome to Startup Web Server</h1>" | sudo tee /var/www/html/index.html`

6. Hardening: Configure the system firewall to permit traffic only on ports 22 and 80.

Ans :- CentOS (Firewalld)

- `sudo firewall-cmd --permanent --add-service=ssh`

- `sudo firewall-cmd --permanent --add-service=http`
- `sudo firewall-cmd --reload`
- `sudo firewall-cmd --list-all`

7. Automation: Script a cronjob that executes at 02:00 daily to compress /var/www into a .tar.gz archive.

Ans :- Cron Job for Backup

Create script:

- `sudo nano /usr/local/bin/webbackup.sh`

Contents:

- `#!/bin/bash`
- `tar -czf /backup/www-$(date +%F).tar.gz /var/www`

Permissions:

- `sudo chmod +x /usr/local/bin/webbackup.sh`

Add cron:

- `sudo crontab -e`
- `0 2 * * * /usr/local/bin/webbackup.sh`

Section C: Integrated Case Study

Scenario: "CloudScale," a boutique hosting provider, recently performed a kernel update on their primary web server. Upon reboot, the web application is unreachable (502 Gateway Error), and administrators report that SSH login attempts take over 60 seconds to respond. The server is live, but the business is losing revenue due to the downtime and unresponsive management interface.

1. Diagnosis: Which specific log files and system commands would you use to identify why the Apache service failed to start?

Ans :- Diagnose Apache Failure

Check service status:

- `systemctl status apache2`
- `systemctl status httpd`

View logs:

- `journalctl -xe`
- `journalctl -u apache2`
- `journalctl -u httpd`
- `tail -f /var/log/httpd/error_log`
- `tail -f /var/log/apache2/error.log`

Check configuration:

- `apachectl configtest`
- `httpd -t`

2. Connectivity: Analyze why the SSH login is experiencing high latency and suggest a fix related to DNS or authentication settings.

Ans :- Common causes:

DNS reverse lookup delay

Edit:

- `sudo nano /etc/ssh/sshd_config`

Add:

- `UseDNS no`

GSSAPI delay

- `GSSAPIAuthentication no`

Restart:

- `sudo systemctl restart sshd`

3. Recovery: Outline the step-by-step CLI process to manually restart the failed services and verify their operational status.

Ans :- Recovery Procedure

Restart services:

- `sudo systemctl restart sshd`
- `sudo systemctl restart apache2`

or

- `sudo systemctl restart httpd`

Verify:

- `systemctl status sshd`
- `systemctl status apache2`
- `ss -tulnp`
- `curl localhost`

4. Preventive Action: Propose a configuration change to ensure the web server automatically attempts to restart if it encounters a runtime failure.

Ans :- Automatic Service Restart

Edit:

- `sudo systemctl edit apache2`

Add:

- `[Service]`
- `Restart=on-failure`
- `RestartSec=5`

Reload:

- `sudo systemctl daemon-reload`
- `sudo systemctl restart apache2`

5. Security: Justify the use of SSH Key-based authentication over passwords to prevent brute-force attacks during this vulnerable state.

Ans :- Why SSH Keys Are Better Than Passwords

Password Authentication	SSH Key Authentication
Vulnerable to brute-force attacks	Extremely difficult to crack
Users choose weak passwords	Uses strong cryptography
Password reuse risk	Unique key pair
Less secure for automation	Ideal for scripts and DevOps

Therefore, during a vulnerable period after a kernel update, SSH key-based authentication significantly improves server security and reduces the risk of unauthorized access.